

Problem Set 8

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Question 5: How does your estimate compare with the true value of β in (1)?

The OLS estimators of β are extremely close to the true parameter values used to generate the data.

Question 7: Do your answers differ?

The estimated values of β are virtually the same when using nloptr's L-BFGS algorithm. However, the estimates obtained using the Nelder-Mead algorithm are slightly further away from the true values compared to the other methods.

Question 9: In your .tex file, tell me how similar your estimates of $\hat{\beta}$ are to the “ground truth” β used to create the data in (1).

The estimated values $\hat{\beta}$ are very close to the “ground truth” β used to generate the data in (1).

	(1)
X1	1.501 (0.002)
X2	-1.001 (0.002)
X3	-0.252 (0.002)
X4	0.749 (0.002)
X5	3.501 (0.002)
X6	-2.001 (0.002)
X7	0.499 (0.002)
X8	1.003 (0.002)
X9	1.247 (0.002)
X10	2.001 (0.002)
Num.Obs.	100 000
R2	0.991
R2 Adj.	0.991
AIC	145 143.6
BIC	145 248.3
Log.Lik.	-72 560.811
RMSE	0.50